NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L22.4 Literature figure gallery and sentence-level retrieval

Group Idea Lab

Status Candidate for promotion

Period 2026-10-15 17:58 — 2026-10-24 21:41

Stage conclusion

Project report

Literature figure library and sentence-level retrieval

Objective

On top of the `ftkb` full-text library (Zotero 3,149 + EndNote 1,527 + Desktop 12,491 documents, ~600,000 pages), evaluate whether two capabilities are worth promoting into platform tools: (A) a figure library (extracting figures + captions across the entire corpus for caption-text and visual-semantic retrieval); and (B) whether sentence-level indexing + clustering is clearly superior to page-level retrieval. Deliver a one-sentence conclusion for each of A and B by 10-21. Task brief (approved by the PI on 10-15).

Current status

10-17 13:0x: **A/B conclusions reached** (ahead of 10-21): A figure-caption retrieval is worth turning into a platform tool (96% pairing accuracy, English figure search 19/20; add a Chinese translation path and single-column crop fixes before productization); A2 image-to-image search will not be pursued (5/10); B sentence-level clustering will not be pursued (underperformed page-level retrieval on all three questions for structural reasons). B full-corpus encoding was stopped at 10-17 16:18 (172/469 shards; shards and logs retained). Handing A over to the platform for integration constitutes promotion, and the PI has been asked. Conclusion.

Settled decisions

- 2026-10-15 Node established, gated on prerequisite platform update →
- 2026-10-15 Restrict to preparation (scripts + trial run + frozen question bank) while prerequisite platform update is pending; figure-finding questions authored and frozen by the lead →

Settled decisions (continued)

- 2026-10-15 A1 v1 pairing accuracy below threshold; revise three rule categories and rerun →
- 2026-10-15 Revised version passes pairing at 92%, fix 76% cropping completeness first, move both libraries to desktop processing →
- 2026-10-15 Desktop read-only fault; pause A1 pending reboot →
- 2026-10-15 Desktop not rebooted; detour by writing to the Windows drive →
- 2026-10-16 Desktop rebooted and recovered with data intact; switch back to writing on root drive →
- 2026-10-16 A1 preparation complete, waiting only on prerequisite platform update; ask the PI whether to launch early →
- 2026-10-16 Prerequisite platform update went live, A1 full run launched →
- 2026-10-16 A1 decision: pairing 96% and English 19/20 passed, Chinese 12/20 failed; dispatched A2 →
- 2026-10-17 A2 did not pass the bar; image-to-image search will not be pursued →
- 2026-10-17 A/B conclusion: pursue A, do not pursue A2 or B →

Criteria (pre-specified in the brief)

Item	Criterion
A1 caption index	≥90% pairing accuracy on 50 random figure–caption pairs; top-5 hit on ≥14/20 figure-finding questions
A2 image embeddings	Given 10 of our own simulation figures to find reference figures in the literature, top-5 contains a relevant figure on ≥6/10 queries (executed only if A1 passes)
B sentence-level + clustering	Across 3 questions (climate-change wind-speed trend + 2 EVA questions), compare representative sentences + source tables per cluster against `pages` page-level results; the lead judges whether it is clearly better

Trials and results

Date	What was done	One-line result
10-15	EVA line provided 2 mixed-climate questions for B	Impact of separate vs. combined fitting on long-return-period design wind speeds; misclassification and missed-observation bias
10-15	A1 v1 trial run on 400 documents (`ideas`) + lead judgment on 50 pairs	Pipeline works end-to-end; strict caption match 19/40, relaxed 35/40, 10 table titles, 7 false captions, 16 full-page crops; decoding failed on 120/400
10-17	B sentence-level fast path on three questions (`ideas`) + lead verdict	Clustering silhouette 0.07–0.11, representative sentences mostly reference entries; all three questions underperformed page-level top 10
10-17	A2 image embeddings (`ideas`, CLIP ViT-B/32) + lead verdict	387k figures encoded in 2.3 h; top-5 across 10 queries: image-to-image 5/10, text-to-image 3/10, below the ≥6/10 bar
10-16	A1 full-scale evaluation (the lead)	50-pair matching 48/50=96%, crop completeness 34/50; 20 queries top-5: English 19/20, Chinese 12/20

Date	What was done	One-line result
10-16	A1 full-scale figure extraction (`ideas`, desktop 12 cores)	17,167/17,167 items processed, 386,894 figures; 3,998 items had partial skipped figures, 0 whole-document failures; compact index 116 MB (<200 MB), thumbnails ~11 GB; caption vector encoding running
10-16	A1 post-reboot continuation (`ideas`) + lead judgment on v3 20 pairs	All 4,676 PDFs staged; Zotero v3 rerun finished; 20-pair check scored 20/20 pairing and 18/20 (90%) cropping completeness
10-15	A1 pre-full-run preparation (`ideas`, v3)	v3 cropping rules implemented; Desktop 200 + EndNote 50 re-rendered; 2,314/4,676 PDFs staged across the two libraries; interrupted at 23:1x when desktop turned read-only
10-15	A1 revised rerun on 400 documents (`ideas`) + lead judgment on new 50 pairs	Pairing 46/50=92% (passed); complete crop 38/50; 1 false caption, 0 table titles, 0 mid-run aborts across 400 documents; kind split: raster 36%[local path omitted] 22%/`page_region` 42%
10-15	The lead authored and froze 20 figure-finding questions	Covers downbursts, Cp, typhoon profiles, gust factors, spectra, bluff bodies, response control, extreme-value standards, etc.

Parameter source table

Parameter	Value	Source or "assumed"	Where checked
Thumbnail size	Long edge ≤512 px JPEG	Assumed (sufficient to read plot types and axes while keeping storage manageable)	No literature basis
Caption block detection	Starts with Fig.[local path omitted] + number, not followed by predicate verbs like shows/gives, short block, adjacent to figure	Assumed (inductively refined from the 50-pair error analysis)	Trial runs in this line
Full-corpus sentence count	~15 million sentences	Task brief estimate (600,000 pages × ~25 sentences)	Pending empirical update from B

Open items and next steps

- Prerequisite platform update go-live time → lead follow-up check at 10-16 08:30; once live and desktop OCR is complete, notify `ideas` to launch full figure extraction (≤16 cores).
- A1 trial report and 50-pair sample → the lead inspects images to judge pairing rate; after the full run, the lead runs the 20 questions to score hit rate.
- B dispatched to `ideas` (10-16 12:5x, desktop ≤8 cores, due 10-20) → the lead compares against page-level baseline to judge whether it is "clearly better"; share results with the EVA line.
- Dispatch A2 only after A1 passes; the lead selects 10 in-house simulation figures from public figures of the CFD line and one other line.

Literature basis

- Jian Zhang and Jia Shao, *Restructuring Higher Education Curricula, Textbooks, and Pedagogy with Small Language Models* (materials provided by the PI): source of the sentence-level corpus + unsupervised clustering methodology and scripts; figure library retrieval originates from the accompanying slide deck (no code included in the package).
- No other literature basis; based on this line's data.

Notes from the PI

10-17 17:3x review and platform handoff

- Review: 5/5 conditional approval → handed over to the platform; see the review verdict and the 8 launch conditions.
- Correction: `fig-06` and `fig-08` in English were revised to *X* (the top 4 results for `fig-06` are downburst papers); under the strict criterion the English score is 16–17/20, and 19/20 is superseded.
- Next steps: once the platform fixes the index, arrange a re-evaluation on 20 new questions authored and scored by someone other than the lead (≥14/20 in both Chinese and English) + re-sample 50 pairs.
- 10-17 platform reply: item opened and will be accepted against the 8 conditions in the verdict; restricted to the Idea Lab until Condition 6 re-evaluation passes; scheduling will be forwarded after assessment in the platform session.
- 10-18 Condition 6 re-evaluation in progress: (1) 50 pairs judged by the lead, with pairing at 46/50=92% (passed) and cropping completeness at 42/50=84%; (2) 20 new questions independently authored by Reviewer E (given only the tool description without seeing the index), to be frozen upon return and handed to `ideas` to run `figs` and score strictly against caption + bibliographic record + thumbnail.
- 10-18 00:36 The 20 new questions have been received and frozen (authored by Reviewer E, sha256 verified) and handed to `ideas` to run and score (≥14/20 in both Chinese and English).

- 10-18 00:5x In the first pass, `ideas` declined to inspect thumbnails per line 29 of its spec and returned "0/20 unconfirmed" (not a performance score). Updated division of labor: `ideas` assigns A/B/C from caption + bibliographic metadata and compiles the image-inspection queue, while the lead inspects each image for final confirmation (the lead is already inspecting the 50 pairs under the same criterion).
- 10-18 Condition 6 re-evaluation completed: pairing passed at 46/50=92%; on the 20 new questions, Chinese scored 15/20 and English scored 13/20, so English fell short of the pre-specified criterion → not opened center-wide for now, remaining in trial use within our lab; the issue list has been handed over to the platform for fixes, to be re-tested on another new question set (decision).

10-19 pause summary and planning (PI directive of 10-19: pause projects that have no plan)

Status: On hold. No new tasks will be dispatched; until the PI approves, wake-ups are only logged. Running/queued delegations: None (checked at 10-19 08:49; all 8 delegations in the subtree have been returned). The issue list already handed to the platform was a 10-18 notification and does not occupy a delegation slot.

① What was done, key numbers, what did not work, and cost

Item	Result
A1 caption index: full-corpus figure extraction	17,167 PDFs → 386,894 figures, 167,331 with captions (43%); workstation compact index 116 MB + vectors 69 MB, thumbnails 9.9 GB on desktop
A1 legacy index scoring	Pairing 48/50=96%; English figure search 19/20 (relaxed, strict 16–17), Chinese 12/20; complete cropping 68%
A review submission	5/5 conditional approval, 8 launch conditions
A handed to platform	Live as figs (desktop bibliographic backfill, 36k deduplicated entries, 84k thumbnails re-cropped, ~3,100 noisy captions cleaned, Chinese-to-English primary query path); restricted to our lab during the trial
Condition 6 re-evaluation	50-pair matching passed at 46/50=92% (interval 81–97%), cropping 84%; 20 new questions authored by someone other than the lead: **Chinese 15/20, English 13/20, English fell short by one question** → not yet opened center-wide
A2 image-to-image search (CLIP)	10 in-house figures hit 5/10 and 3/10, below ≥6/10 bar; will not be pursued

Item	Result
B sentence-level index + clustering	Underperformed page-level retrieval on all three questions (clusters by topic rather than stance, reference entries drown out conclusion sentences); will not be pursued; encoding stopped at 172/469
By-products	Wind engineering LLM pitfall checklist v2 (20 real incidents, awaiting the PI's ✓/X/? marks); L22.5 plan to distill Kareem's expert judgment submitted to the root to open a node

What did not work: A2 and B; Chinese queries failed the bar on the legacy index (subsequently improved via the platform's Chinese-to-English translation path, scoring 15/20 in Chinese upon re-evaluation); ~1/4 of v4 re-cropped figures had issues; English query ranking was weak, with zero hits across three categories (PV wind pressure / LES vortex structures / multi-country code comparisons). Cost: Colab CU 0; paid API 0 (Gemini covered by subscription quota; question authoring and reviews drawn from the review quota pool); supercomputer SU 0; local desktop compute (A1 full run 4.6 h, A2 encoding 2.3 h, B encoding). My own mistake: In the 10-17 legacy blind test, I scored solely by reading captions without inspecting images, misjudging a downburst paper as a typhoon paper; the 19/20 score is voided.

② If continued

Objective (verifiable): On another new frozen question set (authored by someone other than the lead), figs achieves ≥14/20 in both Chinese and English (strict criterion) and ≥90% on a 20-pair re-sample, after which it is opened center-wide; after opening, evaluate whether to retain it based on 2 weeks of real usage logs (query volume, click-through rate). **Near term (2 weeks)**: ① Week 1: The platform fixes issues per the checklist (re-cropping defective v4 figures, checking whether the three zero-hit categories reflect corpus gaps or ranking issues, and improving English ranking), while I only perform visual re-evaluation; ② Week 2: Re-test on another new set of 20 questions; if it passes, request the platform to enable `open_to_all`; if it fails, run one more revision round with a hard cap. **Medium term (1–3 months)**: Combine the figure library with page-level retrieval `pages` (returning both pages and figures for the same query); backfill captionless figures (OCR pages and scanned pages, currently absent from the index); if valuable, merge caption-level duplicate content into literature cards. **Stop-loss**: If it still fails the bar after one more revision round (English or Chinese <14/20) → put on hold, keeping the current trial with no further investment; if 2 weeks of usage logs show near-zero usage → put on hold. **Items for the PI to decide (recommended options)**: ① Whether to continue A (recommended: continue for one more round, as the platform item is already open, low-cost, and right at the 14-question threshold); ② Formally drop and archive A2 and B (recommended: drop, as the evidence is clear); ③ Mark ✓/X/? on Pitfall Checklist v2 (recommended: mark first, then distribute across research lines).

10-21 summary and future report

- Stop category: On hold (stuck, awaiting the PI; re-test pending platform fixes). Under the 10-18 rule, the PI decides the disposition of A2 and B (recommendation: stop both).

- Report: (supersedes the final section of 10-19 as the external version).
- 10-21 00:5x The PI approved resuming (original objective unchanged). Requested improvement ideas from Reviewers A/E/C and a second frozen question set from Reviewer B; once the ideas return, they will be consolidated and handed to the platform in a single batch. decision
- 10-21 00:54 Second question set received and frozen: , sha256 … (authored by Reviewer B, 20 questions across 17 sub-topics); withheld from both the platform and the idea-generating reviewers, to be run only after platform fixes are complete.
- 10-21 Reviewer A's ideas returned (unutilized English full-text candidates, cross-work prefix deduplication, term and relational evidence, thumbnail quality status, source caption truncation; noted that the corpus does contain relevant items for the three zero-hit categories); waiting for E and C before consolidating for the platform.
- 10-21 Reviewer C's ideas returned: primary cause is near-zero recall of MiniLM on technical discriminative terms (correct figures ranked 300–96,000, outside the top-200 candidate pool); recommended BM25 sparse recall + dense RRF fusion (reusing the unenabled bm25_ranks/fts_en), fixing en_terms, and tagging/down-weighting thumbnails via geometric QC; multi-country code comparison queries may genuinely lack figures in the corpus. Waiting on E.
- 10-21 Ideas from all three reviewers received (consensus: ranking/recall issue rather than missing figures; the lead verified that bm25_ranks is unenabled, fts_en searches captions only, and the corpus contains 37 solar panel / 19 Q-criterion / 108 isosurface / 23 gust-factor–turbulence-intensity entries, vs. only 1 multi-country code comparison entry). Consolidated task brief handed to the platform (with acceptance checklist).
- 10-21 The platform took on the round-2 improvements (six acceptance deliverables to be forwarded on delivery); the lead scheduled a 2-day fallback follow-up.
- 10-23 01:3x Follow-up: the platform delivered the round-2 work at 10-21 02:05 (figkb_v2), plus a minor patch to v2.1 (live at 10-21 03:10), but the delivery was not forwarded here. The lead's visual check on the re-sampled pairs: live v2.1 scored 20/20 and the platform sample scored 19/20 (the single failed entry was already removed in v2.1), totaling 39/40=97.5% (passed). Dispatched `ideas` to run the frozen second question set and perform text-based tiering, with final visual scoring by the lead upon return.
- 10-23 01:4x Round 2 acceptance: On the new questions, Chinese scored 16/20, English scored 15/20, and pairing scored 39/40; all passed → requested the platform to open access center-wide (decision). Primary node objective accomplished; status: "Exploring · Candidate for promotion — awaiting PI review"; disposition of A2 and B still awaits the PI.
- 10-23 01:44 Center-wide access live (config open_to_all=true); Line 30 "Finding figures in the literature" added to the shared rules (including all known limitations, verified by the lead). The platform is investigating why the round-2 deliveries were not forwarded back.

Key figures